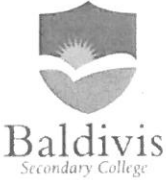
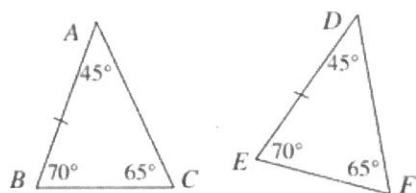


Name:	<u>Anne Sims</u>		Date:	_____
Teacher:	<u>Legendary</u>			
	Year 8 Mathematics Mini test 8 Topic – Congruency		SCORE: / 22	
	<u>Full working out MUST be shown to get full marks for each question.</u>			
Total Time:	30 minutes			
Weighting:	5%			
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, A4 page of notes.			

1. [4 marks]

The following are pairs of congruent triangles

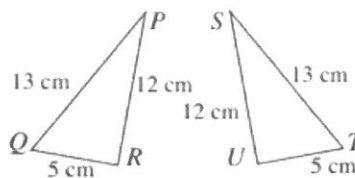


a) Which side corresponds to AC?

DF

b) Write the congruency statement

$\triangle ABC \equiv \triangle DEF$



a) Which angle corresponds to angle QRP?

$\angle TUS$

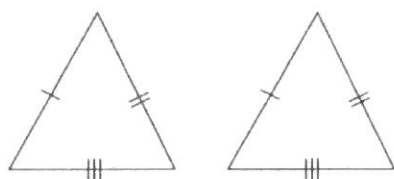
b) Write the congruency statement.

$\triangle QRP \equiv \triangle TUS$

2. [4 marks]

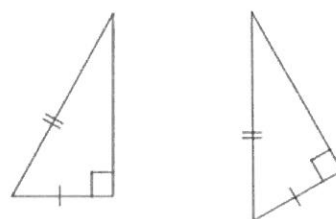
Write the congruency test for each of the following pairs of congruent triangles

a



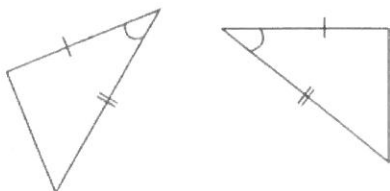
SSS

b



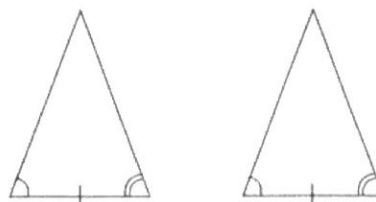
RHS

c



SAS

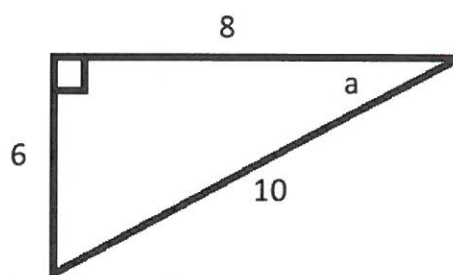
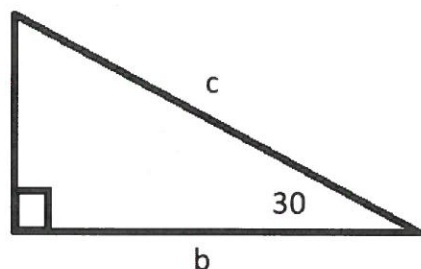
d



AAS

3. [3 marks]

If the triangles are congruent determine the unknown values for a, b and c



two possible answers

✓ $a = 30$

$a = 60$

✓ $b = 8$ or

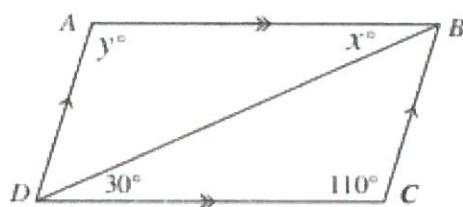
$b = 6$

✓ $c = 10$

$c = 10$

4. [4 marks]

Consider the following parallelogram



a) What is the value of x?

30°

b) What is the value of y?

110°

c) Write the congruency statement for the 2 triangles

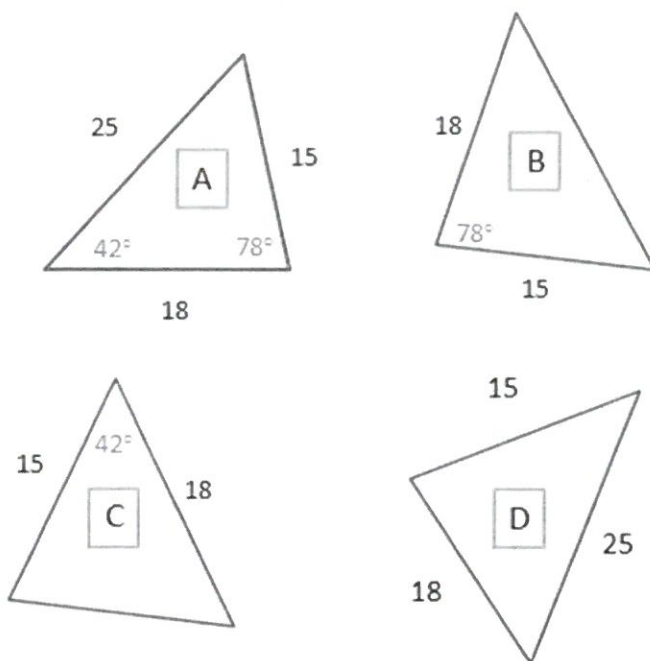
$\triangle ABD \equiv \triangle CBD$

d) Under what transformation will map one triangle to the other?

Rotation 180°

5. [3 marks]

Which of the following triangles are congruent? Why?

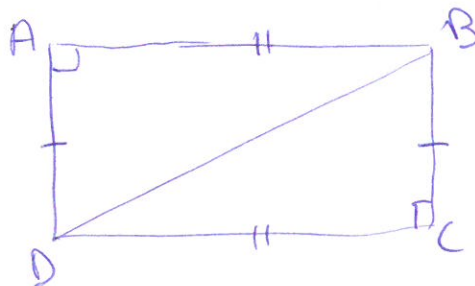


A and B SAS

A and D SSS

6. [4 marks]

When a rectangle has a diagonal drawn across it, it will form two triangles. Explain why these two triangles are congruent.



- ✓ - diagram
- ✓ - Sides/angles
- ✓ - congruency statement
- ✓ - Rule.

$$\begin{aligned}
 AB &= CD & \text{or } \angle DAB &= \angle DCB = 90^\circ \\
 AD &= CB & DB &= DB = \text{Hyp} \\
 DB &= DB & AB &= CD
 \end{aligned}$$

$$\therefore \triangle ABD \equiv \triangle CDB$$

SSS or RHS

END OF TEST

